

based developments, services, and products are all about.

Whatever is being designed today is getting outdated in no time. For instance, if you had developed excellent software for a fax machine, you may need to follow up with software for a cordless machine, and then perhaps for a mobile phone. So whether you like it or not, your software is going to be destroyed and decommissioned sooner or later. The bottom line is that change is constant, and we have to be ready for it.

The key ingredients

If viewed as a recipe, there are three key ingredients needed to build good software.

Skill: Recruiters in India are having a tough time hiring people with the right skills. On the flip side, over 80 per cent of engineers are unemployed mainly because of the lack of new age technology skills. At the same time, the attrition rate for experienced people is now extremely high. The pandemic has created a huge demand for experienced talent, and employers are finding it hard to retain such people.

Long notice periods create additional problems as they give a candidate time to look around, get a job that pays better, and then come back and negotiate with his or her current employer. With a shortage of talent, the employer very often gives in, and offers a higher salary to retain the employee.

Then there is a proliferation of 3-month courses that offer placements, where there is no skill assessment involved. After the course, you suddenly become a geek because of the job you got, irrespective of what you have been taught. However, how far you succeed in this job is anyone's guess.

A few companies have set up their own colleges and hire these students after they graduate, so that their remuneration can be less. Even though

How can India be a powerhouse of open source?

- India has a large pool of developers. However, its contribution to open source is abysmally low for various reasons. Open source is not charity, and companies can ask their people to focus on it whenever it will be helpful. This is definitely not a waste of time, as there will be an outcome.
- Meetups should be considered for recruitment.
- Open source should be part of the college syllabus. All projects in engineering colleges can be developed on open source.
- Government welfare schemes can be made available through open source communities. Quality solutions can be provided without a lot of investment.

this may be a good strategy from a business standpoint, the graduates may not be able to fit in the outside world.

Mindset: Let's take an example to understand this better. Vijay, a school kid extremely interested in computers and an experienced Linux admin, will not be able to find a job in the industry because he is still at school. Divya, an experienced quality analyst, even if extremely talented, will find it difficult to find a job once she takes a career break for personal reasons. Sekar, who is an excellent designer but cannot communicate well, may not find a perfect job as well. An organisation with the right mindset will gladly hire all three individuals and give them the right training.

The technical decisions made by a team will solely depend on the mindset of its members, and the future of any project depends on that.

Purpose: Each software design has a purpose, but the process of digital transformation can have a few challenges.

- Developers should not be disturbed while they are working. However, there is a delay in a lot of releases and, very often, teams are internally divided among themselves.
- If the progress of the project is slow and deadlines are not met, the team still works extra hours, based

on the need. But these extra hours come with a cost.

- There is some internal resistance from the design team itself when the same people end up working together always.
- If technical decisions are made by non-technical people, it can lead to confusion.

Is open source perfect?

When it comes to choosing between writing your own application programming interface (API) and taking one from open source, as much as it entirely depends on the use case, open API is generally used for its security, good quality code, feature availability, extensibility, etc. So, is open source perfect in all areas? Does it have no disadvantages? Not at all. There should be a perfect balance between interfaces and open source, in order to make good software.

But first, the positives...

- **Simplicity:** The skillsets involved are simple and don't demand too many machines, infrastructure, etc. Whatever is available is used.
- **Attracts talent:** Open source attracts talent. Only the passionate opt for it.
- **Developers are important people:** Decisions in the open source domain are mostly made by developers.
- **Remote work:** Work in the open source domain is done remotely, which is a great boon during this